

# Comparison of Speech Dysfluency Types

|                         | Audible Pattern                           | Example                       | Key Acoustic Features                                            |
|-------------------------|-------------------------------------------|-------------------------------|------------------------------------------------------------------|
| Partial Word Repetition | Rapid repetition of first sound           | b-b-b-ball                    | Repeated short bursts (100–300 ms), similar waveform pattern     |
| Whole Word Repetition   | Repetition of full syllable               | ba-ba-baby                    | Repeated syllable energy blocks, clear vowel repetition          |
| Sound Prolongation      | Stretched sound                           | sssssssnake                   | Long sustained frequency band, extended phoneme duration         |
| Pre-utterance Silence   | Sudden stop before word (silent or tense) | ---ball                       | Silence >500 ms or strained low-energy airflow before word onset |
| Interjections           | Inserted filler word/sound                | um... I think                 | Stable mid-frequency tone, nasal vowel patterns                  |
| Phrase Repetition       | Repetition of multiple words              | I went to—I went to the store | Repeated long waveform segment with similar prosody              |
| Rapid Speech            | Very fast, unclear speech                 | iwenttothestore               | High syllable rate, reduced vowel duration, irregular rhythm     |
| Fluent Speech           | Smooth continuous flow                    | I went to the store           | Stable rhythm, natural pauses, no abnormal repetitions           |